AutoML Analytics Pipeline

Automated Data Analysis and Machine Learning Report

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1. Project Overview

This project implements an Automated Machine Learning (AutoML) Analytics Pipeline using a customer and sales dataset. It automates data cleaning, feature engineering, model training, model evaluation, optimization and business visualization.

2. Dataset Information

Parameter	Value
Total Records	1000
Total Columns	18
Total Customers	1000
Missing Values After Cleaning	0

3. Business Performance

Business Metric	Value
Total Sales	7,004,335.87
Total Profit	1,211,118.94
Average Sales	7,004.34
Unique Customers	1000

4. Feature Engineering

The pipeline generated the following additional features:
- Age_Group: Groups customers according to age.
- Customer_Type: Identifies new and returning customers.
- Gross_Order_Value: Quantity multiplied by Unit Price.
- Discount_Amount: Calculates the discount value.
- Profit_Margin: Calculates profit as a percentage of sales.

5. Sales Prediction Model Results

Model	MAE	RMSE	R2 Score
Gradient Boosting	171.0589	347.1129	0.9984
Random Forest	236.3695	484.5069	0.9969
Linear Regression	470.7009	696.6581	0.9937

6. Profit Prediction Model Results

Model	MAE	RMSE	R2 Score
Gradient Boosting	374.1766	650.5885	0.8572

Model	MAE	RMSE	R2 Score
Random Forest	376.8996	653.6295	0.8559
Linear Regression	408.6396	698.2605	0.8356

7. Optimized Sales Model Results

Target	Best Model	MAE	RMSE	R2 Score
Sales	GradientBoostingRegressor	158.6861	326.5875	0.9986

8. Optimized Profit Model Results

Target	Best Model	MAE	RMSE	R2 Score
Profit	RandomForestRegressor	373.8864	646.5846	0.859

9. Business Insights

1. Highest sales category: Electronics.
2. Most profitable city: Aurangabad.
3. Total sales: 7,004,335.87.
4. Total profit: 1,211,118.94.
5. Best Sales Model: Gradient Boosting.
6. Best Profit Model: Gradient Boosting.

10. Conclusion

The AutoML Analytics Pipeline demonstrates an automated workflow for customer and sales data analysis. It performs preprocessing, feature engineering, regression model comparison, optimization and business insight generation. The reported model metrics are experimental results on the available dataset and should be validated on additional data before real-world deployment.